

Introduction to Probability and Statistics

June 2025

Read carefully the exam problems, noting that those with a higher degree of difficulty have been marked with asterisks “***”.

Problem P.1

In a computer network data packets are transmitted. Each data packet consists of 3 fields: a header and two payload fields. The probability to receive a header correctly is $p_1 = 95\%$. The probability to receive the first payload field correctly is $p_2 = 80\%$ and for the second it is $p_3 = 90\%$. The probability to receive parts of a packet are independent of each other. In order to receive the whole packet correctly, all three fields should be received correctly.

Now we know that a sequence of $n = 20$ packets has been sent.

- 1) What is the probability that exactly 18 packets are received correctly?
- 2) What is the probability that at least 18 packets are received correctly?
- 3) What is the expected number of packets that are successfully delivered?
- 4) *** Calculate the probability that at least 3 packets are received correctly for the following set of parameters: $p_1 = 10^{-2}$, $p_2 = 10^{-2}$, $p_3 = 10^{-1}$ and $n = 10^5$.

Problem P.2

The joint probability density function for a random variable (RV) X and a RV Y is given by the following formula:

$$f_{X,Y}(x,y) = \begin{cases} c(1-y), & 0 \leq x \leq 1, \quad 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- 1) Find the constant c .
- 2) Find the marginal probability density functions $f_X(x)$ and $f_Y(y)$.
- 3) Are random variables X and Y independent? Justify your answer.
- 4) Calculate the following probabilities:
 - a) $P(X = 0.5, Y = 0.5)$
 - b) $P(X = 0.5)$
 - c) $P(-1 < X < 0.5)$
 - d) $P(-1 < X < 0.5 \mid Y < 0.5)$
- 5) Find the expectation and variance for X and for Y and their covariance.
- 6) Find the cumulative distribution functions for X and Y .
- 7) *** Now a joint pdf is given by the following formula:

$$f_{X,Y}(x,y) = \begin{cases} 6(1-y), & 0 \leq x \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Draw the area where this function is not zero. Calculate the marginal probability density functions $f_X(x)$ and $f_Y(y)$.

Problem P.3

A mobile robot uses a proximity sensor to detect obstacles in its path. The sensor isn't perfect, and its behavior is characterized as follows:

- 1) If an obstacle is present, the sensor correctly detects it with probability 0.99.
- 2) If no obstacle is present, the sensor still falsely detects one with probability 0.2.

From the geometry of the room and frequency of sampling we know that probability that there is an obstacle in front of the robot is equal to 0.3.

- 1) What is the probability that the sensor detects an obstacle in any given reading?
- 2) The sensor signals that there is an obstacle ahead. What is the probability that there is actually an obstacle?

Problem S.1

An Olympic athlete, who specializes in high jump, has recorded the number of attempts needed to jump above 2 meters during each training day. This is RV X . Her records from the last 8 days, with X_i being the number of attempts until successfully jumping above 2 meters in the i -th day (i.e., experiment), including the successful attempt, are: 5, 4, 4, 3, 4, 2, 1, 3. The records are shown in Table I for clarity.

TABLE I
RECORDS FOR AN OLYMPIC ATHLETE THROUGH 8 DAYS.

Day	1st attempt	2nd attempt	3rd attempt	4th attempt	5th attempt
1	✗	✗	✗	✗	✓
2	✗	✗	✗	✓	
3	✗	✗	✗	✓	
4	✗	✗	✓		
5	✗	✗	✗	✓	
6	✗	✓			
7	✓				
8	✗	✗	✓		

- 1) Estimate $\bar{X}_n$, which is the average number of attempts until successfully jumping above 2 meters.
- 2) Calculate the two-sided 94% confidence interval for $\bar{X}_n$ using the t-distribution.
- 3) *** Let $k \in \{0, 1, 2, \dots\}$ be the number of failed attempts until a successful jump for each individual day. Then, RV $X \sim \text{Geo}(p)$ with probability mass function (pmf)

$$\Pr(X = k + 1) = (1 - p)^k p$$

Obtain the Maximum Likelihood Estimator (MLE) for parameter p , denoted as $\hat{p}_n$. Include the formula for the likelihood or log-likelihood function, depending on which one you use to calculate $\hat{p}_n$, as well as your estimate for the provided data.

Problem S.2

The evolution of 3 different phenomena through time has been recorded through several days. The values obtained for each of them are shown in Table II.

TABLE II
VALUES RECORDED FOR THREE PHENOMENA ACROSS SEVERAL DAYS.

Day	Phenomenon 1	Phenomenon 2	Phenomenon 3
1	1	0	0.2
2	3.8	2.4	11.5
3	0	5.0	12.8
4	13.9	10.3	22.0
5	6.8	16.1	21.9
8	22.4	18.5	25.2
9	31.7	22.4	29.7

- 1) Obtain the regression coefficients for each of the phenomena.
- 2) Determine for which of the three phenomena the linear model has a better fit. Provide the name and formulation of the coefficient that you used to support your answer and the values obtained for the three phenomena.

Problem S.3

A specific type of light bulb is advertised to produce 1600 lumens. To test this product, the following luminosity measurements, given in lumens, were collected:

1550, 1620, 1490, 1599, 1580, 1601

Make the following tests with 2% significance for the mean luminosity of the light bulbs μ . Provide your test statistic and rejection region or the p-value to support your conclusion.

- 1) A two-sided test for the null hypothesis $H_0: \mu = 1600$. Provide the estimated mean $\hat{\mu}_n$ and sample standard deviation S_n .
- 2) A test for the null-hypothesis $H_0: \mu > 1600$.

*** Now consider that the sample size is increased to $n = 20$ but both the estimated mean $\hat{\mu}_n$ and the sample standard deviation S_n remain the same.

- 3) What is the minimum level of significance to reject the null hypothesis $H_0: \mu = 1600$? Is it rejected with $n = 20$?
- 4) What is the minimum level of significance to reject the null hypothesis $H_0: \mu > 1600$? Is it rejected with $n = 20$?